

Assignment 2 - Linear Algebra

Each original question is reproduced below, followed immediately by its answer.

Part A - Scalars, Vectors and Matrices

Question 1

Question: Identify whether each of the following is a Scalar, Vector, Matrix, or Tensor.

- a) 5
- b) $[2, 4, 6]^T$
- c) $[[1, 2], [3, 4]]$
- d) A colour image of size $256 \times 256 \times 3$

Answer: a) Scalar.
b) Vector (3×1).
c) Matrix (2×2).
d) Tensor (a 3-dimensional array).

Question 2

Question: Determine the dimensions of the following matrices.

- a) $[[1, 2, 3], [4, 5, 6]]$
- b) $[5, 8, 9, 2]^T$
- c) $[[1, 2], [3, 4], [5, 6]]$

Answer: Dimensions are rows x columns.
a) 2×3 .
b) 4×1 .
c) 3×2 .

Question 3

Question: Given $A = [[2, 5], [3, 7]]$ and $B = [[4, 1], [6, 8]]$, find:

- a) $A + B$
- b) $A - B$
- c) $2A$

Answer: a) $A+B = [[2+4, 5+1], [3+6, 7+8]] = [[6, 6], [9, 15]]$.
b) $A-B = [[2-4, 5-1], [3-6, 7-8]] = [[-2, 4], [-3, -1]]$.
c) $2A = [[4, 10], [6, 14]]$.

Question 4

Question: Find the transpose of the matrix $A = [[2, 4, 6], [8, 10, 12]]$.

Answer: Transposing swaps rows and columns: $A^T = [[2, 8], [4, 10], [6, 12]]$.

Question 5

Question: Find the transpose of $[[1, 3], [2, 5], [4, 6]]$.

Answer: Transposing swaps rows and columns: $[[1, 2, 4], [3, 5, 6]]$.

Part B - Matrix Multiplication

Question 6

Question: Multiply $A = [[1, 2], [3, 4]]$ and $B = [[5, 6], [7, 8]]$.

Answer: $AB = [[1(5)+2(7), 1(6)+2(8)], [3(5)+4(7), 3(6)+4(8)]] = [[19, 22], [43, 50]]$.

Question 7

Question: Determine whether the following matrix multiplications are possible. If possible, write the dimensions of the resulting matrix.

- a) $(2 \times 3)(3 \times 4)$
- b) $(3 \times 2)(4 \times 3)$
- c) $(4 \times 2)(2 \times 5)$
- d) $(5 \times 4)(4 \times 2)$

Answer: The inner dimensions must match; the result uses the outer dimensions.

- a) Possible; 2×4 .
- b) Not possible because $2 \neq 4$.
- c) Possible; 4×5 .
- d) Possible; 5×2 .

Question 8

Question: Given $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, find AB .

Answer: $AB = \begin{bmatrix} 2(5)+1(7) & 2(6)+1(8) \\ 4(5)+3(7) & 4(6)+3(8) \end{bmatrix} = \begin{bmatrix} 17 & 20 \\ 41 & 48 \end{bmatrix}$.

Question 9

Question: Verify whether matrix multiplication is commutative. Compute AB and BA where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Based on your calculations, what do you conclude?

Answer: $AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$.

$BA = \begin{bmatrix} 5(1)+6(3) & 5(2)+6(4) \\ 7(1)+8(3) & 7(2)+8(4) \end{bmatrix} = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}$.

Since $AB \neq BA$, matrix multiplication is **not commutative in general**.

Part C - Vector Operations

Question 10

Question: Find the magnitude of the vector $x = [3, 4]^T$.

Answer: $|x| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

Question 11

Question: Find the magnitude of $y = [2, 5, 6]^T$.

Answer: $|y| = \sqrt{2^2 + 5^2 + 6^2} = \sqrt{65} = 8.06$.

Question 12

Question: Compute the dot product of $x = [2, 4, 1]^T$ and $y = [5, 2, 3]^T$.

Answer: $x \cdot y = 2(5) + 4(2) + 1(3) = 10 + 8 + 3 = 21$.

Question 13

Question: Determine whether the following vectors are orthogonal: $u = [2, 1]^T$ and $v = [1, -2]^T$.

Answer: $u \cdot v = 2(1) + 1(-2) = 0$. Therefore, **u and v are orthogonal**.

Question 14

Question: Normalize the vector $x = [6, 8]^T$.

Answer: $|x| = \sqrt{6^2 + 8^2} = 10$.

$x/|x| = [6/10, 8/10]^T = [0.60, 0.80]^T$.

Question 15

Question: Find the Euclidean distance between $A = (2, 3)$ and $B = (8, 11)$.

Answer: $d(A,B) = \sqrt{(8-2)^2 + (11-3)^2} = \sqrt{36+64} = \sqrt{100} = 10$.

Part D - Vector Spaces and Span

Question 16

Question: Determine whether the following vectors are linearly independent: $v_1 = [1, 2]^T$ and $v_2 = [2, 4]^T$. Explain your answer.

Answer: $v_2 = 2v_1$, so one vector is a scalar multiple of the other. The vectors are **linearly dependent**.

Question 17

Question: Determine whether $v = [5, 7]^T$ belongs to the span of $v_1 = [1, 1]^T$ and $v_2 = [2, 1]^T$.

Answer: Solve $a+2b=5$ and $a+b=7$. Subtracting gives $b=-2$; then $a=9$.

Thus $v = 9v_1 - 2v_2$, so **yes, v belongs to the span**.

Question 18

Question: Find a basis for $\mathbb{R}^2$.

Answer: $\{[1, 0]^T, [0, 1]^T\}$ is a basis: these vectors are linearly independent and span $\mathbb{R}^2$.

Question 19

Question: Find the rank of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

Answer: The second row is twice the first, so there is one independent row. **Rank = 1**.

Question 20

Question: Find the rank of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Answer: The determinant is $1(4)-2(3) = -2$, which is nonzero. **Rank = 2**.

Part E - Orthogonality

Question 21

Question: State whether the following pairs of vectors are orthogonal.

- a) $[1, 0]$, $[0, 1]$
- b) $[3, 4]$, $[4, -3]$
- c) $[2, 5]$, $[5, 2]$

Answer: a) Dot product = 0: **orthogonal**.

b) Dot product = $3(4)+4(-3) = 0$: **orthogonal**.

c) Dot product = $2(5)+5(2) = 20$: **not orthogonal**.

Question 22

Question: Find the cosine similarity between $x = [1, 2]^T$ and $y = [3, 4]^T$. Interpret the result.

Answer: $\cos(\theta) = (x \cdot y)/(|x||y|) = 11/(\sqrt{5} \times 5) = 11/(5\sqrt{5}) = 0.98$ (to two decimals).

The vectors point in very similar directions.

Question 23

Question: Explain, in your own words, the difference between orthogonal vectors, orthonormal vectors, a basis, and a vector space.

Answer: Orthogonal vectors: vectors with dot product zero.

Orthonormal vectors: mutually orthogonal vectors, each with magnitude one.

A basis: a linearly independent set that spans the space.

A vector space: a set closed under vector addition and scalar multiplication and satisfying the vector-space axioms.